# EDS Assignment

## PS50008A Research Methods | Term 2 Spring 2026

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## Key Information

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|---|---|
| **Deadline** | Friday 20 March 2026, 12 noon |
| **Submission** | Via Moodle submission link |
| **Format** | Worked statistics assignment |
| **Weight** | 20% |
| **Pass mark** | 50% |

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## Task Structure

You will receive a research scenario with **pre-generated R output**. Your task is to interpret the output and demonstrate your understanding of the statistical analysis - no coding or hand calculations required.

### Section A: Statistical Mechanics (50 marks)

1. State the researcher's hypothesis (5 marks)
2. Identify design type, IV, DV, data type (16-20 marks)
3. **Interpret R descriptive output** and report in APA format (10 marks)
4. **Interpret R t.test() output** and justify test choice (15 marks)
5. Full APA results paragraph from R output (5 marks)
6. **NEW: Error detection** - identify what's wrong with a given analysis (5 marks)

### Section B: Methods and Theory (50 marks)

7. Suggest TWO methodological improvements
8. Discuss individual differences / extensions
9. Propose a more ecologically valid alternative (minimum half page)

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## Critical Requirements

> ⚠ Demonstrate Understanding
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> You MUST demonstrate your understanding - not just get the right answer.
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> - Full marks = right test AND proper justification
> - No punitive marking, so if you get something wrong, you just get zero for that item
> - You are expected to use the terminology and concepts we have been discussing in class
> - Pay attention to the glossaries and readings provided

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## What "Demonstrate Understanding" Means

| Aspect | What we're looking for |
|---|---|
| **Test selection** | Why this test? Link to design and data type |
| **Hypothesis** | Clear statement with direction justified |
| **Reading output** | Correctly identify key values (t, df, p, CI) |
| **Interpretation** | What does the result mean in context? |
| **Error detection** | Identify when an analysis doesn't match the design |
| **Improvements** | Thoughtful, specific, feasible suggestions |

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## Preparation

- Practice interpreting R output from lab exercises (weeks 12-16)
- Review APA reporting format for statistics
- Understand the logic behind test selection
- Be comfortable reading and interpreting R output
- Practice error detection exercises (what's wrong with this analysis?)

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## Example R Output You'll Encounter

### Descriptive Statistics

```
> data %>%
+   group_by(condition) %>%
+   summarise(Mean = mean(RT), SD = sd(RT), n = n())

# A tibble: 2 x 4
```

```
  condition      Mean    SD     n
  <chr>         <dbl> <dbl> <int>
1 30_degree      285.  42.3    10
2 150_degree     312.  51.1    10
```

## T-Test Output

```
> t.test(data$RT_30, data$RT_150, paired = TRUE)

    Paired t-test

data:  data$RT_30 and data$RT_150
t = -2.847, df = 9, p-value = 0.0192
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -49.82 -5.38
sample estimates:
mean of the differences
                  -27.6
```

You will NOT need to write R code. Your task is to:

1. Read the output correctly
2. Explain what the numbers mean
3. Report findings in APA format
4. Detect when an analysis contains errors